

Customer Churn Analysis Dashboard - Project Report

This report presents a comprehensive Customer Churn Analysis Dashboard developed in Power BI. The objective of this project is to identify customer churn patterns, understand retention drivers, quantify revenue loss, and provide actionable business recommendations to improve customer retention and profitability.

Project Overview & Business Objectives

Introduces the customer churn problem, project objectives, dashboard navigation, key metrics, tools used, and expected business outcomes.

Customer Churn Analysis Dashboard

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Problem Statement:

The objective of this project is to analyze the customer churn behavior and identify the key factors contributing to customer attrition. By leveraging data visualization techniques, this dashboard aims to help businesses understand churn patterns and take proactive actions to improve customer retention.

Dashboard Created:

1.Customer Churn Analysis & Retention

Insights - Worked on KPIs like Total Customers, Churn Rate, Retention Rate, Revenue loss and gathered insights like **~18%** revenue loss due to Contract type and Partner Status and recommending to reduce monthly charges and strategies for retention improvement.

2.Customer Segmentation & Churn

Behavior Insights - Worked on KPIs like Churn rate, Total tenure, Avg. monthly charges which leads to insights that **~2M** Revenue loss due to monthly plans, recommending to reduce the monthly charges for increasing the total tenure on Monthly Basis.

Key Metrics Preview

- 1.Total Customers
- 2.Churned Customers
- 3.Churn Rate %

Tool Used

- 1.Power Bi
- 2.DAX
- 3.Data Cleaning

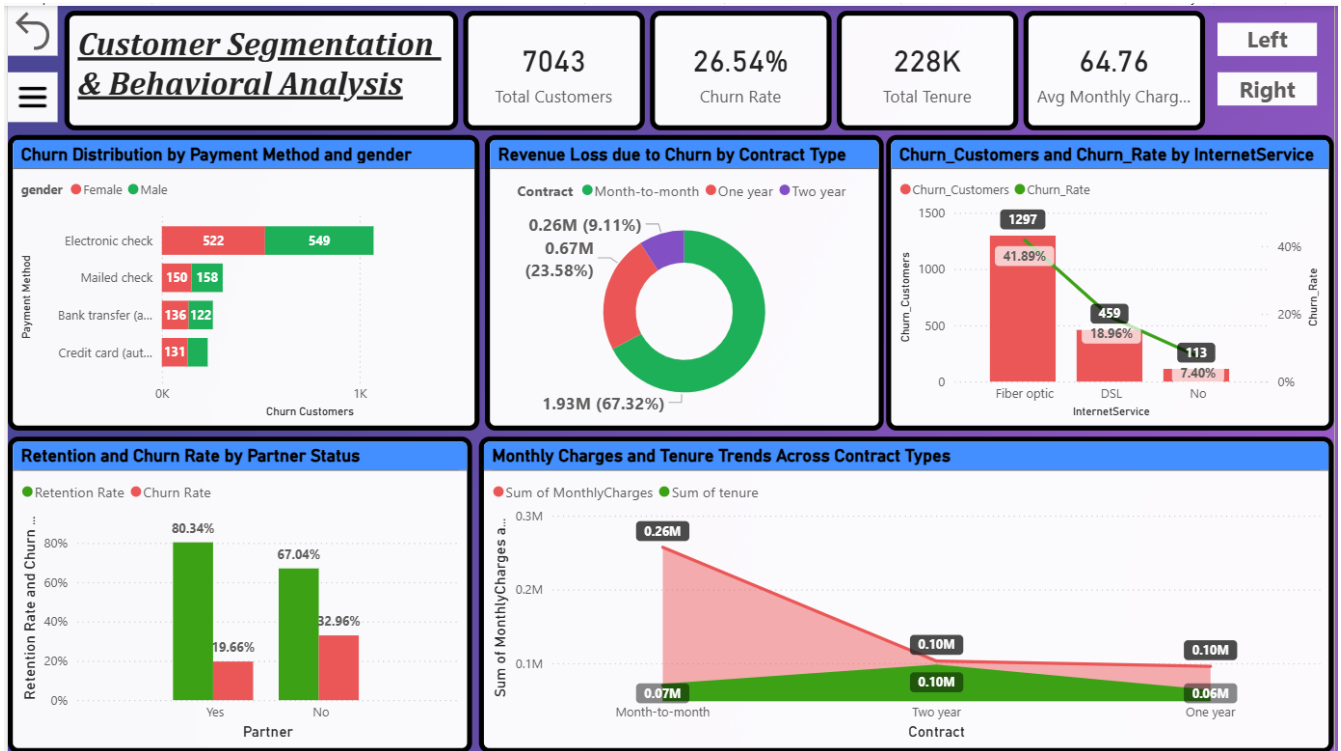
Customer Churn & Retention Analysis

Analyzes churn rate, retention rate, customer counts, revenue loss, contract type behavior, payment methods, device protection impact, and revenue contribution.



Customer Segmentation & Behavioral Analysis

Examines customer behavior across payment methods, contract types, internet services, partner status, monthly charges, and tenure patterns.



Key Insights

Highlights major findings including high churn among month-to-month customers, revenue loss concentration, internet service impacts, payment method behavior, and customer profile trends.

<u>Key Insights:</u>	Left Right
	1.Contract Type Insights: 1.Month-to-Month contracts contributing to highest customer churn (~42.7%), indicating low customer commitment and higher risk of revenue loss. 2.Revenue Type Insights: 1.Approximately ~67% of total revenue loss is driven by month-to-month customers, making them the most critical segment for retention strategies. 2.Over all approximately ~18% total revenue loss due to different contract types. 3.Payment Methods Insights: 1.Bank Transfer, Credit Card, and Electronic Check has the highest impact of payment methods on customer tenure Approximately (~0.20M). 2.Customers using electronic check shows the higher churn compared to other payment methods, indicating that payment friction may influence customer retention. 4.Internet Service Insights: 1.Fiber optic users exhibits the highest churn rate (~42%), suggesting potential dissatisfaction despite being the high-value customers. 5.Add-On Service Insights: 1.Customers with device protection have significantly higher retention rate (~85-95%), highlighting the importance of value-added services in the reducing churn. 6.Customer Profile Insight: 1.Customers Without partners show a higher churn rate (~33%) compared to partnered customers (~19%), indicating that single users are more likely to discontinue the services. 7.Pricing Insights: 1.Customers with higher monthly charges and lower tenure are more likely to churn, suggesting a mismatch between pricing and value.

Business Recommendations

Provides strategic recommendations focused on retention improvement, automated payments, service quality enhancement, pricing optimization, and long-term customer engagement.

Business Recommendations:

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1. 🎯 Encourage Long-Term Contracts:

Introduce discounts, loyalty rewards, and bundled offers for yearly and multi-year plans to reduce churn from month-to-month customers.

2. 💳 Promote Automated Payment Methods:

Encourage customers to shift from electronic checks to auto-pay methods through cashback offers and payment incentives to improve retention.

3. 🌐 Improve Fiber Optic Service Experience:

Focus on service quality, customer support, and pricing optimization for fiber optic users, as they represent the highest churn-risk customer segment.

4. 📦 Increase Adoption of Value-Added Services:

Bundle device protection and additional support services with internet plans to improve customer retention and long-term engagement.

5. 🏷️ Optimize Pricing for High-Charge Customers:

Reassess pricing strategies and provide personalized offers for customers with high monthly charges to reduce dissatisfaction and churn risk.

Executive Summary

Key findings reveal that month-to-month customers contribute the highest churn rate and revenue loss. Customers using electronic checks and fiber optic services exhibit elevated churn behavior. Device protection, longer contract durations, and partner relationships positively influence retention. Strategic initiatives such as promoting long-term contracts, encouraging automated payments, improving service quality, and optimizing pricing can significantly reduce churn and increase customer lifetime value.